Keng qamrovli tarmoq asoslari

Wide Area Network Fundamentals

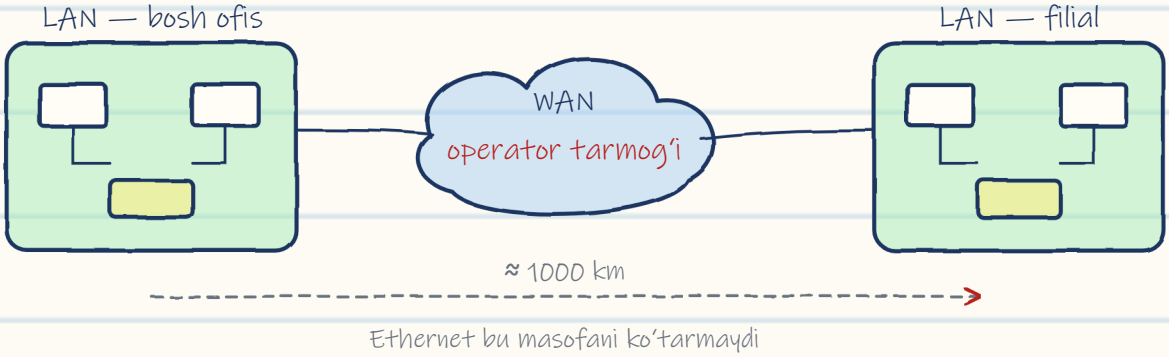
Asosiy atamalar

- WAN — shahar, viloyat, davlat oralig'ini bog'laydigan tarmoq
- DTE / DCE — mijoz qurilmasi / operator qurilmasi, taktni DCE beradi
- Ajratilgan liniya — doimiy, nuqta-nuqta, qat'iy tezlikdagi kanal
- Kanal kommutatsiyasi — yo'l terib ochiladi, tugagach uziladi
- Paket kommutatsiyasi — har paket manzili bilan alohida ketadi
- Virtual kanal (VC) — bitta fizik portda ko'p mantiqiy ulanish

1. Nega WAN kerak?

LAN — bino, kampus. WAN shahar, viloyat, davlat chegarasidan o'tadi

- 100BASE-TX — 100 m, repeater bilan 500 m, kommutator bilan bir necha km
- o'z liniyasi — qimmat, ruxsat yo'q → operator tarmog'i ijaraga



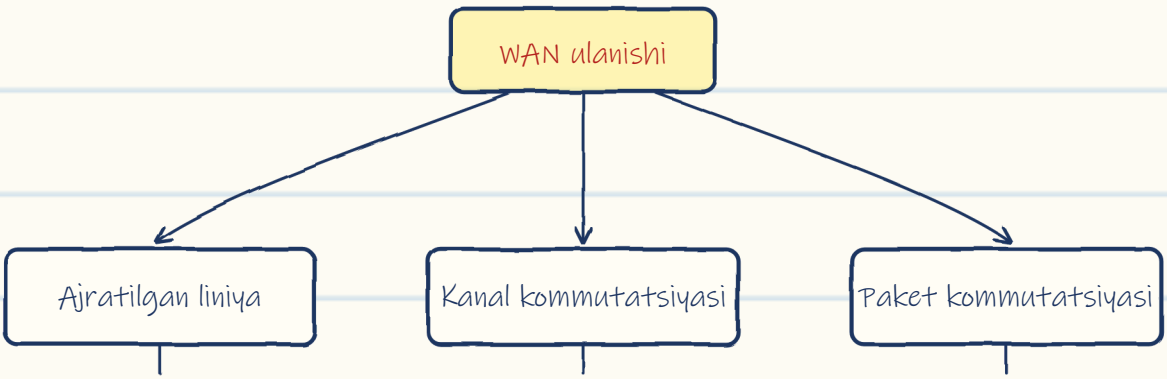
1-rasm. Uzoqdagi LAN'lar operator tarmog'i orqali bog'lanadi

2. WAN va OSI modeli

OSI sathi	WAN da nima ishlaydi
Tarmoq sathi	IP · IPX
Kanal sathi	HDLC · PPP · LAPB Frame Relay
Fizik sath	V.24 · V.35 · X.21 RS-232 · RS-449 · RS-530 G.703 — E1 (Yevropa/Osiyo) · T1 (Amerika)

WAN = kanal + fizik sath

3. Uch xil WAN ulanishi



WAN ni qurish uchun router kerak

4. Ajratilgan liniya (leased line)

Mijozga doimiy, nuqta-nuqta, qat'iy tezlikdagi o'z liniyasi.



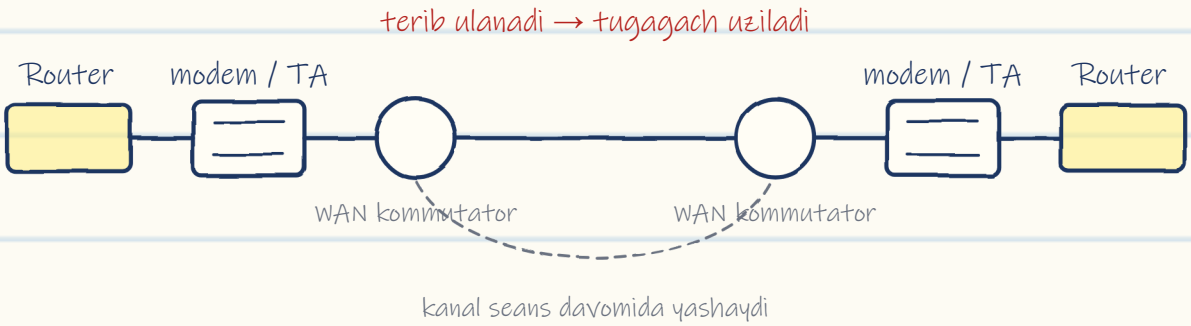
DCE takt beradi · DTE taktni oladi

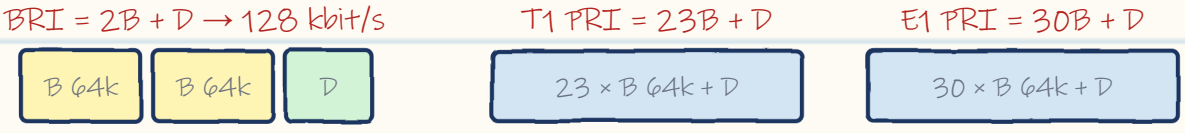
tezlikni operator belgilaydi, polosa to'liq mijozniki

2-rasm. Kanal sathi: SDLC · HDLC · PPP

5. Kanal kommutatsiyasi

Kommutator yo'lni faqat kerak bo'lganda ochadi, tugagach uzadi.





PSTN: analog, maks. 56 kbit/s, shovqinga sezgir

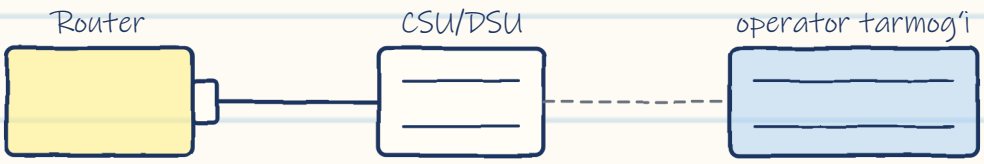
ISDN: D — signalizatsiya, B — ma'lumot

- PSTN — arzon, hamma binoda bor; polosa kichik
- ISDN — tez, ishonchli; 2 ta B → 2 ta ilova

Interfeys	BRI	T1 PRI	E1 PRI
B kanal	2 × 64 kbit/s	23	30
D kanal	1	1	1

Qoida: kanal kommutatsiyasi — vaqtinchalik, kichik polosali aloqa uchun; ulanish kechikadi.

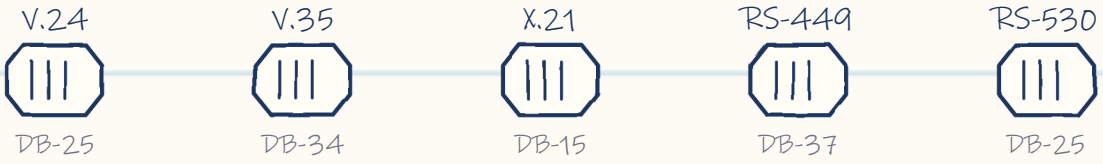
6. Interfeys va kabellar



serial kabel · 28-pinli ulagich

kirish kabeli — juftlik, RJ-11 / RJ-45

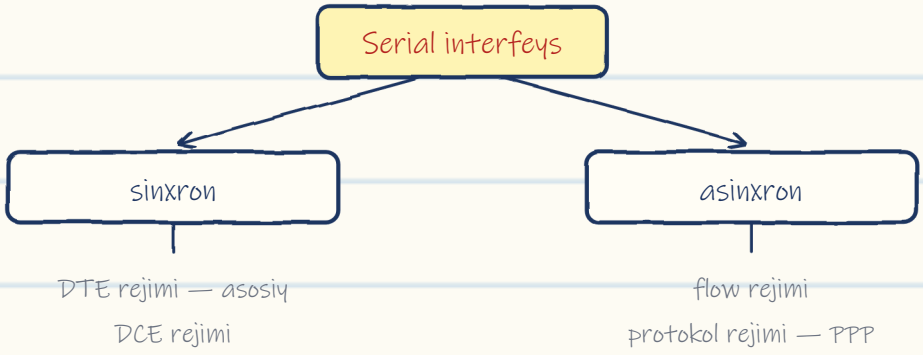
tarmoq tomondagi ulagichlar



DTE kabel — erkak (pin) · DCE — urg'ochi (teshik)

	V.24 (RS-232)	V.35
Rejim	sinxron + asinxron	faqat sinxron
Maks. tezlik	64 000 bit/s sinx · 115 200 bit/s asinx	2 048 000 bit/s (2 Mbit/s)
9600 bit/s da	30 m	312 m

X.21 · RS-449 · RS-530 — tezlik va masofasi V.35 ga yaqin, hammasi 2 Mbit/s.

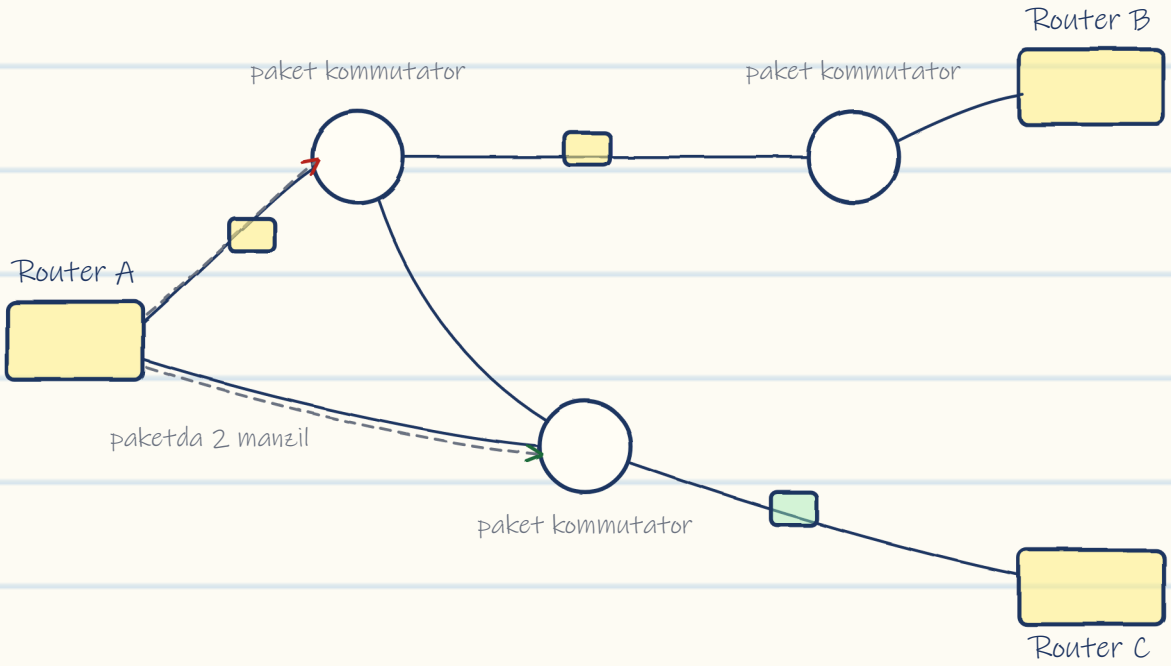


asinxron — modem yoki ISDN TA bilan dialup

7. Paket kommutatsiyasi

Paketlarga bo'linadi, har paketda jo'natuvchi va qabul qiluvchi manzili.

virtual kanal (VC) — bitta fizik port, ko'p peer



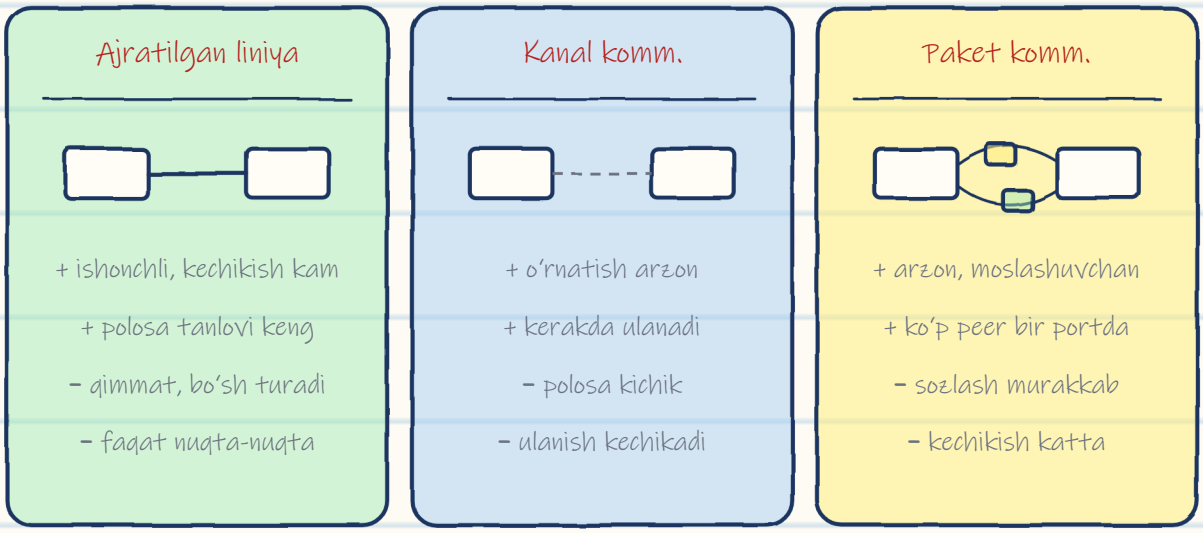
X.25 · Frame Relay · ATM

sinxron ajratilgan liniya bilan ulanadi

	X.25	Frame Relay	ATM
Tezlik	64 kbit/s, 1992-dan 2 Mbit/s	64 kbit/s – 2 Mbit/s	1000 Mbit/s gacha
Xususiyat	xato tuzatish, oqim nazorati — sekin	X.25 ning yengili, kechikish kam	hujayra (cell), optika, qimmat

+ / −: moslashuvchan, arzonroq; sozlash murakkab, kechikish katta. Frame Relay tiqilishga sezgir.

8. Uch rejim — taqqoslash



Hammasida: DTE ↔ DCE va kanal sathi protokoli

Xulosa

- WAN — kanal sathi + fizik sath; operator quradi
- Fizik: V.24 · V.35 · X.21 · RS-449 · RS-530 · G.703 (E1/T1)
- Kanal: HDLC · PPP · LAPB · Frame Relay · SLIP · SDLC
- 3 rejim: ajratilgan liniya · kanal · paket kommutatsiyasi
- Router = DTE, CSU/DSU = DCE va taktni beradi

Tekshirish

1. WAN qaysi 2 sathda? — fizik va kanal
2. V.35 maks. tezligi? — 2 Mbit/s, faqat sinxron
3. ISDN BRI? — 2B + D = 128 kbit/s
4. Taktni kim beradi? — DCE (CSU/DSU)
5. Point-to-multipoint qaysi rejim? — paket kommutatsiyasi